National University of Computer and Emerging Sciences

Lahore Campus

Intro to Cloud Computing (CS4037) BDS8A

Sessional-II Exam

Date: April 9th, 2025

Course Instructor(s): Ali Khawaja

Total Time (Hrs.):	**1**
Total Marks:	**100**
Total Questions:	**7**

CLO 1. Apply cloud computing knowledge in various real world use cases (situations)

Q1: How do Tags help in: (10 Points)

a. **Workload Optimization and automation**: Tags can help you visualize all of the resources that participate in complex deployments. For example, you might tag a resource with its associated workload or application name and use software such as Azure DevOps to perform automated tasks on those resources.

b. **Governance & regulatory compliance**: Tags enable you to identify resources that align with governance or regulatory compliance requirements, such as ISO 27001. Tags can also be part of your standards enforcement efforts. For example, you might require that all resources be tagged with an owner or department name.

c. **How can enforce tagging rules to be applied during creation of resources?** Through Azure Policy

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Q2: Resource Locks (20 Points)

a. **What is the purpose of resource locks?** Protect your Azure resources from accidental deletion or modification.

b. Briefly describe the type of resource locks available in Azure

   a. Delete Lock
   b. Readonly Lock

c. **How can you modify a locked resource?** To modify a locked resource, you must first remove the lock. After you remove the lock, you can apply any action you have permissions to perform.

d. **How can an administrator disallow modification of resources once they are properly configured?** Through ReadOnly Lock

e. **How can an administrator disallow deletion of resources while allowing any updates?** Through Delete Lock

CLO 1. Describe the basics of cloud computing and related services

Q3: Azure Arc (10 Points)

a. **Briefly and concisely provide the purpose of Azure Arc**: In utilizing Azure Resource Manager (ARM), Arc lets you extend your Azure compliance and monitoring to your hybrid and multi-cloud configurations.

b. Name three resource types that can be managed by Azure Arc (***Any 3 of the below***)

   a. Servers
   b. Kubernetes clusters

   c. Azure data services
   d. SQL Server
   e. Virtual machines (preview)

c. Briefly list 2-3 services provided by Azure Arc (***Any 2 of the below***)

   a. Manage your entire environment together by projecting your existing non-Azure resources into ARM.
   b. Manage multi-cloud and hybrid virtual machines, Kubernetes clusters, and databases as if they are running in Azure.
   c. Use familiar Azure services and management capabilities, regardless of where they live.
   d. Continue using traditional ITOps while introducing DevOps practices to support new cloud and native patterns in your environment.
   e. Configure custom locations as an abstraction layer on top of Azure Arc-enabled Kubernetes clusters and cluster extensions.

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Q4: Azure Resource Manager (10 Points)

a. **Briefly describe ARM**: Azure Resource Manager (ARM) is the deployment and management service for Azure. It provides a management layer that enables you to create, update, and delete resources in your Azure account. Anytime you do anything with your Azure resources, ARM is involved.
b. Mention 2 benefits of ARM Templates. (***Any 2 of the below***)

   a. **Declarative syntax**: ARM templates allow you to create and deploy an entire Azure infrastructure declaratively. Declarative syntax means you declare what you want to deploy but don't need to write the actual programming commands and sequence to deploy the resources.
   b. **Repeatable results**: Repeatedly deploy your infrastructure throughout the development lifecycle and have confidence your resources are deployed in a consistent manner. You can use the same ARM template to deploy multiple dev/test environments, knowing that all the environments are the same.
   c. **Orchestration**: You don't have to worry about the complexities of ordering operations. Azure Resource Manager orchestrates the deployment of interdependent resources, so they're created in the correct order. When possible, Azure Resource Manager deploys resources in parallel, so your deployments finish faster than serial deployments. You deploy the template through one command, rather than through multiple imperative commands.
   d. **Modular files**: You can break your templates into smaller, reusable components and link them together at deployment time. You can also nest one template inside another template. For example, you could create a template for a VM stack, and then nest that template inside of templates that deploy entire environments, and that VM stack will consistently be deployed in each of the environment templates.
   e. **Extensibility**: With deployment scripts, you can add PowerShell or Bash scripts to your templates. The deployment scripts extend your ability to set up resources

during deployment. A script can be included in the template or stored in an external source and referenced in the template. Deployment scripts give you the ability to complete your end-to-end environment setup in a single ARM template.

c. **What is the difference between bicep file and an ARM template?** Bicep files tend to use a simpler, more concise style.

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Q5: Monitoring in Azure (20)

a. **Which is the key monitoring platform in Azure?** Azure Monitor

b. **What is the role of Azure Log Analytics in this monitoring platform?** Azure Log Analytics is the tool in the Azure portal where you'll write and run log queries on the data gathered by Azure Monitor.

c. **Which functionality allows an administrator to be notified of any issue being detected?** Azure Monitor Alerts

d. **What feature allows you to take corrective action once an issue is detected?** Action Group

e. **For application monitoring in Azure environments, mention some of the information that is available through application insights.**

    a. Request rates, response times, and failure rates
    b. Dependency rates, response times, and failure rates, to show whether external services are slowing down performance
    c. Page views and load performance reported by users' browsers
    d. AJAX calls from web pages, including rates, response times, and failure rates
    e. User and session counts
    f. Performance counters from Windows or Linux server machines, such as CPU, memory, and network usage

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Q6: Azure Advisor (10)

a. **What is the purpose of azure advisor?** Azure Advisor evaluates your Azure resources and makes recommendations to help improve reliability, security, and performance, achieve operational excellence, and reduce costs.

b. Name any 3 categories of recommendations by Azure Advisor along with brief explanations. (***Any 3 of the below***)

    a. **Reliability** is used to ensure and improve the continuity of your business-critical applications.

b. **Security** is used to detect threats and vulnerabilities that might lead to security breaches.
c. **Performance** is used to improve the speed of your applications.
d. **Operational Excellence** is used to help you achieve process and workflow efficiency, resource manageability, and deployment best practices.
e. **Cost** is used to optimize and reduce your overall Azure spending.

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Q7: Azure Service Health. Briefly explain the following 3 services: (20)

a. **Azure Status:** informs you of service outages in Azure on the Azure Status page. The page is a global view of the health of all Azure services across all Azure regions.

b. **Service Health:** provides a narrower view of Azure services and regions. It focuses on the Azure services and regions you're using. This is the best place to look for service impacting communications about outages, planned maintenance activities, and other health advisories because the authenticated Service Health experience knows which services and resources you currently use.

c. **Resource Health**: is a tailored view of your actual Azure resources. It provides information about the health of your individual cloud resources, such as a specific virtual machine instance. Using Azure Monitor, you can also configure alerts to notify you of availability changes to your cloud resources.